

PHASE 23A – TRAINING SOURCE OF TRUTH AUDIT V1
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Repo: /Users/walterelrafie/macro-workout-coach

Scope

- Audit only.
- No new features.
- No UI changes.
- No canonical engine changes.
- No Training Planner changes.

Repo state observed before PDF generation

Modified:

- src/app/page.tsx
- src/lib/home-command-center.test.ts
- src/lib/home-command-center.ts

Untracked:

- TRAINING_PLANNER_ARCHITECTURE_V1.md
- src/lib/training-planner.test.ts
- src/lib/training-planner.ts

EXECUTIVE SUMMARY

The application currently has multiple competing training sources of truth:

1. src/lib/training-planner.ts
 - New canonical Phase 22 source.
 - Not wired yet.
2. src/app/page.tsx
 - Independently computes today's workout, today's run, planned distance, scheduled run, daily prescription, completion state, and Train display state.
3. src/lib/home-command-center.ts
 - Independently derives today's run, run distance/duration, Home training model, Home completion status, and calls legacy evaluateTraining().
4. src/lib/training-engine.ts
 - Legacy session composer.
 - Creates warmup/workout/run/cooldown blocks, status, duration, warnings, priorities.
 - Overlaps heavily with DailyTrainingSession.
5. src/lib/coach-engine.ts
 - Legacy recommendation generator.
 - Creates daily workout prescription, running recommendation, readiness-adjusted workout, next-set and post-workout recommendations.
6. src/lib/seed-data.ts
 - Current plan data and fallback workout selector.
 - Contains hardcoded 12-week schedule, run/mobility prescriptions embedded as workout exercises, and fallback to workouts[0].
7. src/app/page.tsx Train UI + run logger
 - Uses TrainingEngineResult, scheduledRun, and plannedRunDistance.
 - Contains defaulting logic that can turn run duration/distance assumptions into logging defaults.

Observed bug paths

- Repeated "Hold: 3 mi": page.tsx passes staleRunLabel: "Hold: 3 mi" into getTodayRunForDate(), then nextRunLabel and getScheduledRunForTraining() can carry the

t label forward.

- Home and Train disagreeing: Home receives `homeCommandCenter.trainingEngineResult`, while Train also has an internal fallback `evaluateTraining()` call with different fake/default progression and goal inputs.
- Run duration becoming reps: seed workouts encode "35-45 min" as `Exercise.prescribedReps`; legacy engines parse/filter this text differently.
- Mobility interpreted as workouts: seed data encodes mobility as exercises; legacy lift detection often means anything not run/walk/sprint/interval.
- Duplicate logging/completion behavior: Train creates workout sessions directly in `page.tsx`; run logging is separate via `TrainRunLogger` -> `saveTrainRunLog()`; completion is independently derived by Home, Train, Daily Check-In, and Weekly Review.

CATEGORY A – MUST BE MIGRATED

These are current sources that generate, transform, infer, or display daily training and should eventually consume or derive from `DailyTrainingSession`.

1. `src/app/page.tsx` – `Home()` daily derivation block, approx lines 146-176
Purpose: Computes today's date mapping, today's workout, adjusted workout, today's run, planned run distance, running recommendation, next run label, scheduled run, daily prescription, and Home command center model.
Current sources: `getWorkoutForWeekDay()`, `getTodayRunForDate()`, `generateRunningRecommendation()`, `getScheduledRunForTraining()`, `generateDailyPrescription()`, `adjustWorkoutForReadiness()`, `buildHomeCommandCenter()`.
Consumes: `state.currentWeek`, `selectedWeek`, `todayIso()`, `workouts`, latest check-in/readiness, run logs/trends, macro targets, post-workout recommendations.
Produces: `todayPlanDayIndex`, `todayWorkout`, `adjustedTodayWorkout`, `todayRunDisplay`, `plannedRunDistance`, `runningRecommendation`, `nextRunLabel`, `scheduledRunForTraining`, `dailyPrescription`, `homeCommandCenter`.
Replace with: `DailyTrainingSession`.
Priority: P0.
2. `src/app/page.tsx` – `todayPlanDayIndex`, approx lines 148-151
Purpose: Maps date to Monday=0/Sunday=6.
Current source: local inline date mapping.
Replace with: `DailyTrainingSession.dayIndex`.
Priority: P0.
3. `src/app/page.tsx` – `todayWorkout`, approx line 155
Purpose: Selects today's workout from `state.currentWeek` + `todayPlanDayIndex`.
Current source: `getWorkoutForWeekDay()`.
Replace with: `DailyTrainingSession.sourcePlan` + `DailyTrainingSession.workout`.
Priority: P0.
4. `src/app/page.tsx` – `adjustedTodayWorkout`, approx line 157
Purpose: Applies readiness modification to today's workout.
Current source: `adjustWorkoutForReadiness()`.
Replace with: `DailyTrainingSession.status`, `modifications`, `workout`, `recovery`.
Priority: P0.
5. `src/app/page.tsx` – `todayRunDisplay`, approx line 170
Purpose: Determines whether today has a run and creates display object.
Current source: `getTodayRunForDate()`.
Consumes `staleRunLabel`: "Hold: 3 mi".
Replace with: `DailyTrainingSession.run`.
Priority: P0.
6. `src/app/page.tsx` – `plannedRunDistance`, approx line 171
Purpose: Converts today's run display into numeric distance.
Current source: `todayRunDisplay?.distanceMiles ?? 0`.
Replace with: `DailyTrainingSession.run.loggingTarget.plannedDistance`.

Priority: P0.

7. src/app/page.tsx – runningRecommendation, approx line 172

Purpose: Generates run recommendation based on planned distance and run logs.

Current source: generateRunningRecommendation().

Replace with: DailyTrainingSession.run plus advisory engine input.

Priority: P0/P1 depending on adapter design.

8. src/app/page.tsx – nextRunLabel, approx line 173

Purpose: Creates displayed run label, including Hold/Progress/etc.

Current source: todayRunDisplay + runningRecommendation.

Replace with: DailyTrainingSession.run.title or summary.runName.

Priority: P0.

9. src/app/page.tsx – scheduledRunForTraining, approx line 174

Purpose: Converts run display/recommendation into scheduled run for Training Engine.

Current source: getScheduledRunForTraining().

Replace with: DailyTrainingSession.run.loggingTarget.

Priority: P0.

10. src/app/page.tsx – dailyPrescription, approx line 175

Purpose: Legacy full daily coach prescription.

Current source: generateDailyPrescription().

Replace training-specific fields with DailyTrainingSession; keep nutrition advice separate.

Priority: P0 for training content.

11. src/app/page.tsx – homeCommandCenter, approx line 176

Purpose: Builds Home model and Home training display.

Current source: buildHomeCommandCenter() fed with adjustedTodayWorkout, nextRunLabel, scheduledRunForTraining, dailyPrescription.

Replace with: Home adapter consuming DailyTrainingSession.

Priority: P0.

12. src/app/page.tsx – TrainScreen()/TrainingPlan(), approx lines 337-747

Purpose: Displays and executes current training session.

Current sources: trainingEngineResult, workout prop, scheduledRun, fallback evaluateTraining().

Produces: visible Train blocks, lift/run scheduled flags, workout session start, run logger display, training completion status.

Replace with: DailyTrainingSession.

Priority: P0.

13. src/app/page.tsx – TrainingPlan() fallback evaluateTraining(), approx lines 658-670

Purpose: Creates internal fallback TrainingEngineResult if not provided.

Current source: inline evaluateTraining() call with fake progression/goal tracking objects.

Replace/remove after planner adapter.

Priority: P0.

14. src/app/page.tsx – TrainingPlan() display flags, approx lines 671-681

Purpose: Derives blocks, lift/run scheduled flags, active session, completion status.

Current source: trainingEngineResult.todayPlan/workout/run plus logs.

Replace with: DailyTrainingSession.blocks, loggingTarget, completion adapter.

Priority: P0.

15. src/app/page.tsx – TrainingPlan.startWorkout(), approx lines 688-706

Purpose: Creates WorkoutSession for lift block.

Current source: displayedWorkout.

Replace with: `DailyTrainingSession.workout.loggingTarget`.
Priority: P1.

16. `src/app/page.tsx` – `TrainRunLogger()`, approx lines 912-950
Purpose: Displays run logging form and saves today's run.
Current source: `plannedDistance`, `runType`, `existingRun`.
Issue: defaults duration as `Math.round(plannedDistance * 11)`; duration-only runs cannot be represented correctly.
Replace with: `DailyTrainingSession.run.loggingTarget`.
Priority: P0.

17. `src/lib/home-command-center.ts` – `getTodayRunForDate()`, approx lines 201-223
Purpose: Determines today's run from current week/date/workouts.
Current source: seed workouts + local run detection/parsing.
Replace with: `DailyTrainingSession.run`.
Priority: P0.

18. `src/lib/home-command-center.ts` – `milesFromWorkout()`, approx lines 155-163
Purpose: Extracts run distance from `Workout.longRunMiles` or exercise reps text.
Current source: text parsing from `Workout/Exercise.prescribedReps`.
Replace with: `DailyTrainingSession.run.distanceMiles`.
Priority: P0.

19. `src/lib/home-command-center.ts` – `estimatedMinutesFromWorkout()`, approx lines 165-174
Purpose: Derives run estimated duration from distance or `prescribedReps` regex.
Replace with: `DailyTrainingSession.run.durationMinutes / estimatedDurationMinuteS`.
Priority: P0.

20. `src/lib/home-command-center.ts` – `isRunWorkout()`, approx lines 189-195
Purpose: Classifies workout as run.
Current source: `title/type/exercise` regex.
Replace with: `DailyTrainingSession.run !== null / blocks`.
Priority: P0.

21. `src/lib/home-command-center.ts` – `getScheduledRunForTraining()`, approx lines 225-233
Purpose: Converts `TodayRunDisplay` into Training Engine scheduled run.
Issue: returns null when `distanceMiles <= 0`, which drops duration-only runs.
Replace with: `DailyTrainingSession.run.loggingTarget`.
Priority: P0.

22. `src/lib/home-command-center.ts` – `buildHomeCommandCenter()`, approx lines 388-521
Purpose: Builds Home dashboard training state and calls legacy Training Engine.
Produces: `todaysWorkout`, `todaysRun`, Home training workout/run, durations, statuses, `trainingEngineResult`.
Replace with: Home adapter over `DailyTrainingSession`.
Priority: P0.

23. `src/lib/home-command-center.ts` – `completedWorkoutToday()/completedRunToday()`
Purpose: Infers training completion for Home.
Replace with: `DailyTrainingSession-aware` completion adapter.
Priority: P1.

24. `src/lib/training-engine.ts` – `evaluateTraining()`, approx lines 186-259
Purpose: Legacy session composer.
Consumes: readiness, progression, goal tracking, scheduled workout, scheduled run, preferences.
Produces: today plan blocks, warmup/workout/run/cooldown, order, duration, priorities, warnings, status, audit trail.

Replace with: buildDailyTrainingSession().
Priority: P0.

25. src/lib/training-engine.ts – runDistance(), approx lines 88-92
Purpose: Derives run distance from scheduled run or Workout.longRunMiles.
Replace with: DailyTrainingSession.run.distanceMiles.
Priority: P0.

26. src/lib/training-engine.ts – estimateRunMinutes(), approx lines 105-110
Purpose: Estimates run minutes from distance.
Replace with: DailyTrainingSession.run.durationMinutes / estimatedDurationMinute
S.
Priority: P0.

27. src/lib/training-engine.ts – hasLift()/workout block creation, approx lines 85 and 209-216
Purpose: Classifies and renders workout/lift items.
Issue: mobility can become workout/lift if not excluded.
Replace with: DailyTrainingSession.workout and blocks.
Priority: P0.

28. src/lib/training-engine.ts – buildRecoveryResult(), approx lines 147-168
Purpose: Creates recovery session from readiness/progression.
Replace with: DailyTrainingSession.recovery and blocks.
Priority: P0.

29. src/lib/coach-engine.ts – generateDailyPrescription(), approx lines 746-862
Purpose: Legacy daily prescription for workout/cardio/nutrition/recovery.
Replace training/recovery/cardio fields with DailyTrainingSession; keep nutrition target logic separate.
Priority: P0 for training content.

30. src/lib/coach-engine.ts – generateRunningRecommendation(), approx lines 481-492
Purpose: Generates running recommendation via Running Engine V2 adapter.
Keep as advisory input only; planner should own final DailyTrainingSession.run.
Priority: P1.

31. src/lib/coach-engine.ts – runningEngineResultToLegacyRecommendation(), approx lines 441-479
Purpose: Converts Running Engine action into legacy run distance/message.
Issue: creates Hold/Progress distance labels outside canonical session.
Replace with planner adapter/advisory-only signal.
Priority: P1.

32. src/lib/coach-engine.ts – adjustWorkoutForReadiness(), approx lines 864-873
Purpose: Mutates workout based on readiness; can create Recovery Replacement as workout-shaped object.
Replace with: DailyTrainingSession.status, modifications, recovery.
Priority: P0.

33. src/lib/seed-data.ts – templateForDay(), approx lines 23-42
Purpose: Generates hardcoded 12-week workout schedule.
Current source: static templates with lifts, runs, mobility, recovery stored as exercises.
Keep as source plan data for now; planner must own interpretation.
Priority: P0 for interpretation.

34. src/lib/seed-data.ts – workouts, approx lines 44-65
Purpose: Materializes all week/day workouts.
Keep as plan source input until structured plan import.
Priority: P1.

35. src/lib/seed-data.ts – getWorkoutForWeekDay(), approx lines 155-157
Purpose: Selects workout by week/day with fallback workouts[0].
Issue: fallback can create incorrect training instead of safe unavailable/rest session.
Replace with: DailyTrainingSession.sourcePlan.
Priority: P0.
36. src/lib/run-logger.ts – buildTrainSessionBlocks(), approx lines 231-283
Purpose: Legacy Train block builder from workout/run distance.
Replace with: DailyTrainingSession.blocks.
Priority: P1.
37. src/lib/run-logger.ts – saveTrainRunLog(), approx lines 285-310
Purpose: Saves run log from Train.
Keep save function, but input should come from DailyTrainingSession.run.loggingTarget.
Priority: P1.
38. src/lib/daily-checkin.ts – deriveDailyCompletionStatus(), approx lines 50-57
Purpose: Derives workout/run completion for Daily Check-In.
Replace with: adapter comparing logs against DailyTrainingSession logging target.
Priority: P1.

CATEGORY B – ADAPTER LAYER ONLY

These should not necessarily be deleted, but should become consumers/adapters around DailyTrainingSession.

1. src/lib/home-command-center.ts – buildHomeCommandCenter()
Keep as Home model builder, but training fields should adapt from DailyTrainingSession:
- workout.name <- session.workout?.title
- run.name <- session.run?.title
- estimatedDurationMinutes <- session.estimatedDurationMinutes
- warnings <- session.warnings
- todaysGoals <- session.todayGoals + nutrition goals
- trainingEngineResult should eventually disappear or be replaced by planner-backed adapter output.
2. src/app/page.tsx – Dashboard()
Pure display component. Keep UI; feed it planner-backed Home model.
3. src/lib/home-daily-dashboard.ts
Display model builder. Safe if fed correct Home command center model.
4. src/app/page.tsx – TrainScreen()
Thin wrapper. Keep, but pass DailyTrainingSession.
5. src/app/page.tsx – TrainingPlan()
Keep display/execution UI; change source from TrainingEngineResult to DailyTrainingSession.
6. src/app/page.tsx – TrainRunLogger()
Keep UI; defaults must come from DailyTrainingSession.run.loggingTarget, including duration-only targets.
7. src/lib/run-logger.ts – buildRunLoggerRecord(), saveRunLoggerEntry(), saveTrainRunLog()
Keep save/evaluation utilities. Inputs should be planner logging targets.

8. `src/lib/workout-logger.ts` – `buildWorkoutLoggerSession()`, `saveWorkoutLoggerEntry()`, `evaluateWorkoutLoggerResult()`
Keep persistence/evaluation utilities. Inputs should be planner workout logging targets.
9. `src/app/page.tsx` – `ActiveWorkout()`
Keep set-by-set execution; feed from planner workout prescription/logging target.
10. `src/lib/coach-engine.ts` – `generateNextSetRecommendation()`, `generatePostWorkoutAnalysis()`, `getNextWorkoutStep()`
Keep set-by-set/post-workout execution engines.
11. `src/lib/coach-engine.ts` – `generateRunningRecommendation()`
Keep as advisory input only; do not let it independently create final displayed/scheduled run labels.
12. `src/lib/daily-checkin.ts` – `upsertDailyCheckIn()`
Keep check-in save behavior; completion should come from canonical scheduled-target adapter.
13. `src/lib/weekly-review.ts` – `buildWeeklyReviewSummary()`
Keep as weekly log summary; may later compare logs against planner targets.
14. `src/lib/mvp-dashboard.ts` – `buildMvpDashboard()`
Keep as display model; feed labels from `DailyTrainingSession` adapter.
15. `src/lib/progress-insights-ui.ts`
Keep; may receive planner-aware progression context later.

CATEGORY C – SAFE TO LEAVE ALONE

1. `src/lib/training-planner.ts`
New canonical Phase 22 source. Keep.
2. `src/lib/readiness-engine.ts`
Canonical readiness signal. Keep.
3. `src/lib/progression-engine.ts`
Canonical weekly progression decision. Keep.
4. `src/lib/running-engine.ts`
Running progression/readiness engine. Keep as advisory engine.
5. `src/lib/workout-engine.ts`
Set/workout progression engine. Keep.
6. `src/lib/goal-tracking-engine.ts`
Goal status engine. Keep.
7. `src/lib/performance-engine.ts`
Historical performance trend engine. Keep.
8. `src/lib/physique-engine.ts`
Physique trend engine. Keep.
9. `src/lib/nutrition-engine.ts`
Nutrition decision engine. Keep.
10. `src/lib/weekly-review.ts` – `buildRunningEngineInputForWeeklyReview()`
Weekly running context from logs. Keep.

11. src/lib/race-calendar-engine.ts – evaluateRaceCalendar()
Race countdown/phase/recommended long-run ranges. Keep; not today's executable source.
12. src/lib/adaptive-training-calendar-engine.ts – evaluateAdaptiveTrainingCalendar()
Weekly roadmap targets. Keep; do not let it become daily source of truth.
13. src/lib/race-calendar-ui.ts – buildRaceCalendarUiModel()
Race roadmap/countdown UI model. Keep.
14. src/lib/storage.ts
Persistence/date/ID utilities. Keep.
15. src/lib/backup-restore.ts
Backup/restore safety layer. Keep.
16. src/lib/log-tab-ui.ts
Log tab metadata/display helper. Keep.

MIGRATION MAP BY APP AREA

Home currently depends on:

- page.tsx daily derivation block: todayWorkout, adjustedTodayWorkout, todayRunDisplay, plannedRunDistance, runningRecommendation, nextRunLabel, scheduledRunForTraining, dailyPrescription, homeCommandCenter.
- home-command-center.ts: getTodayRunForDate(), getScheduledRunForTraining(), buildHomeCommandCenter(), legacy evaluateTraining().
- training-engine.ts: evaluateTraining().

Home should eventually depend on:

- buildDailyTrainingSession().
- Home adapter: DailyTrainingSession -> Home command center fields.
- Nutrition/progression/goal engines remain separate where appropriate.

Train currently depends on:

- adjusted legacy workout object.
- scheduled run object.
- trainingEngineResult.
- fallback evaluateTraining().
- direct workout session creation.
- direct run logging defaults.
- run-logger save/evaluation.
- coach-engine set-by-set and post-workout logic.

Train should eventually depend on:

- DailyTrainingSession.blocks.
- DailyTrainingSession.workout.
- DailyTrainingSession.run.
- DailyTrainingSession.warmup.
- DailyTrainingSession.cooldown.
- DailyTrainingSession loggingTarget fields.
- Keep set-by-set workout execution, run save function, post-workout analysis.

Log currently depends on:

- deriveDailyCompletionStatus().
- upsertDailyCheckIn().
- existing run/workout logs indirectly.

Log should eventually depend on:

- Completion adapter: DailyTrainingSession + workout/run logs -> scheduled/completed/missed status.

- Keep Daily Check-In form, nutrition/body/photo logging.

Progress currently depends on:

- calculateRunTrends().
- buildWeeklyReviewSummary().
- buildHomeCommandCenter() with blank workout/run options.
- buildRaceCalendarUiModel().
- buildProgressInsightsModel().

Progress should eventually depend on:

- Mostly keep existing.
- Replace ProgressScreen call to buildHomeCommandCenter(... todaysWorkout: "", todaysRun: "") with planner-backed adapter or non-training engine context builder
- Weekly review may later compare logs to DailyTrainingSession targets.

HIGHEST-RISK LEGACY LOGIC TO REPLACE FIRST

1. src/app/page.tsx
 - todayRunDisplay
 - plannedRunDistance
 - nextRunLabel
 - scheduledRunForTraining
 - dailyPrescription
 - homeCommandCenter
2. src/lib/home-command-center.ts
 - getTodayRunForDate()
 - getScheduledRunForTraining()
 - buildHomeCommandCenter() training section
3. src/lib/training-engine.ts
 - entire evaluateTraining() daily-session composition path
4. src/app/page.tsx
 - TrainingPlan() fallback evaluateTraining()
5. src/app/page.tsx
 - TrainRunLogger() defaults based only on planned distance
6. src/lib/seed-data.ts
 - getWorkoutForWeekDay() fallback to workouts[0]

RECOMMENDED PHASE 23B SCOPE

Phase 23B – Training Planner Adapter V1

Build adapter only. No UI redesign.

Recommended actions:

- Create a read-only adapter that converts current app state into BuildDailyTrainingSessionInput.
- Call buildDailyTrainingSession() once for today.
- Do not remove legacy engines yet.
- Do not change UI layout yet.
- Compare adapter output against legacy Home/Train values in tests.

Required test cases:

- No repeated “Hold: 3 mi”.
- Duration run remains duration-based.
- Mobility does not become lift reps.
- Home and Train receive the same canonical session object.

- Missing workout is safe rest/unavailable, not workouts[0] fallback.

END OF AUDIT